How far ahead of the case count do public data forecast a large dengue season? A retrospective simulation by Brazilian state

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Abstract. Background: the number of Brazilian states with a high-incidence dengue season (300 or more probable cases per 100,000) ranged from 3 to 19 between 2014/15 and 2024/25. It is not known how far ahead of the simple case count public data allow forecasting this. Methods: retrospective simulation in the 27 states and 11 seasons (297 pairs), each state with its own season, at each of the first 43 weeks, partially rebuilding what was available on each date. Eight predictors (count projection, recent growth, off-season cases, mild winter, thermal suitability, El Niño and, by amendment, cases in the capital and in the interior, each compared with its own history) were tested alone and in pairs, in penalized logistic regressions anchored on the state's history; criterion M chooses, each season, the model with the lowest error in earlier seasons. Lead time is the number of weeks between accuracy stably passing 75% and the count of probable cases making half of the high seasons evident. The main analysis uses the final classification of cases; a sensitivity analysis uses all notifications, as an alternative approximation of what was available. Additional analyses, defined after the first results: a fixed alert rule, a three-state rule with the projection band and a test on held-out seasons. Results: the count projection passed 75% accuracy at week 10, 25 weeks before evidence from the count (95% CI 14 to 30; 11 under the strict rule), and 18 weeks (12 under the strict rule) with the count less dependent on the final classification, in seasons 2016/17 to 2020/21. The interior improved the projection in every version (incremental BSS 0.06 to 0.14), more than off-season cases in most analyses and with a very close value in one of them; the capital added no clear gain to the projection. Criterion M, whose final composition is projection plus interior, had 18 weeks (CI 11 to 29; 10 under the strict rule) in the main analysis and 11 (5 under the strict rule) in the version less dependent on the final classification. The fixed pair projection plus interior had the lowest error and the longest lead time under the strict rule (13 weeks), and was the one chosen when selection used only 2011/12 to 2018/19, but those held-out seasons had already been examined. With a probability alert rule, all models alerted before the count in most high seasons, with false alarms in 37% to 57% of non-high seasons; requiring the whole projection band above the threshold cut false alarms to 0% to 2%, but the median lead time to 0 to 4 weeks, and declared low, in some week, 10% to 25% of the seasons that ended high. Conclusion: in this retrospective simulation, the count projection anticipates evidence from the count; with criterion M, whose final composition is the projection and the cases in the interior (in the retrospective forecasts, M changed models over the seasons), a high season can be forecast 18 weeks ahead of the simple case count in the main analysis, but the lead time falls with the count less dependent on the final classification, and false alarms are frequent. Confirmation depends on the prospective check.

Keywords: dengue; forecasting; lead time; predictors; SINAN; Brazilian states; capital and interior.

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https://mais-saude-ai.vercel.app/en/notes/janela-previsao-dengue/
Translated from the Portuguese original.

In plain language

Why did we do this study? Some years have a lot of dengue and others have little. People who run health services in each state need to know early whether the year will be bad. We wanted to know how much spare time the data the government publishes give before simply adding up the cases makes everything obvious.

What did we do and what did we find? We went back over eleven past years and pretended to be in each week, using what was already known. We tested eight clues, such as heat, winter, El Niño, cases that show up outside the usual season and cases in the capital and in the rest of the state. The most useful clue was the running total of cases itself, compared with each state's usual pace. It helped even more to check whether the cities outside the capital already had more cases than they usually do. With this, on average, the answer was right in three out of four forecasts some months before the result became obvious. But part of this advantage came from already knowing which cases would later be ruled out. A simple alert would also raise many alarms for nothing, and a more cautious alert almost never raises alarms for nothing, but comes late and sometimes says the year will be calm when it will not.

What does this mean? Public data already give an early warning, but the test looked back in time. It still has to be seen whether it works in the coming years. This note says nothing about the health of any person.

I. INTRODUCTION

A. The problem and why it matters

Dengue in Brazil is seasonal, but the size of each season varies greatly from one year to the next. Between 2014/15 and 2024/25, the number of states with a high-incidence season

ranged from 3 to 19, out of 27 (Table II). For those who organize beds, staff, supplies and campaigns, the practical question is: how far ahead do public data make it possible to say that this year will be a big one in my state? And is that lead time greater than what the case count itself would already give?

Each week, more data arrive. At the start, they say little; at the end, the case count already shows the result without any model at all. A forecasting model is only useful if it gets it right before that moment. The distance, in weeks, between the point at which the model starts to get it right and the point at which the count makes the result evident is the **lead time** that this note measures. It is not, in itself, the lead time relative to the start of the epidemic or to the peak; for that reason, the note also measures how many high seasons the model anticipates before the peak.

B. What is already known

Forecasting dengue is hard, and the literature is candid about it. In an open challenge with 16 teams, eight seasons and two cities in the Americas, forecasts made at the start of the season had little skill, skill was lowest precisely in the high-incidence seasons, and the models with more biology were not, on average, better than the simple ones [5]. In Vietnam, an ensemble of climate-driven models outperformed a historical baseline at 1 to 3 months ahead, but not at 4 to 6 [6]. In Brazil, seasonal climate forecasts made it possible to issue warnings three months before the 2014 World Cup [3], and the evaluation of that prototype used historical averages as the comparison [4]. The InfoDengue system has estimated the cases not yet reported every week (nowcasting) since 2015 [2]; the InfoDengue-Mosqlimate consortium brings teams together to forecast each state's incidence in the following season [7, 8]; and there are studies that combine official data and internet searches to estimate the current week [9].

These works were located through a general internet search and by reading the full text in Europe PMC, on 7 Nov 2026. There was no systematic search of the literature (PubMed, SciELO, LILACS and other databases were not searched systematically); for that reason, this note does not claim that the approach is new.

C. The question

This note asks, using the aggregated public data from SINAN:

- **Predictors:** which signals, alone and in pairs, improve the forecast of a high-incidence season relative to the state's history, and how much does each one add to the count projection?
- **Start:** from which week of the season does the chosen forecasting criterion get the yes-or-no forecast right in 75% or more of the state-season pairs, without falling below that again?
- **Lead time:** how many weeks separate that start from the week in which the simple case count of the probable cases already available makes half of the high seasons evident?

Why 75% accuracy? It is the level at which the forecast is wrong, on average, in at most one out of every four state-season pairs. It is above the accuracy of someone who always answers "no" (54% in the period, because 46% of the seasons were high) and the accuracy of someone who only looks at the state's history (64% to 66%). It is a usefulness threshold, set in the protocol, and not a significance test. Overall accuracy is not the same as identifying 75% of the high seasons; for that reason the note also reports the sensitivity (fraction of the high seasons identified), the specificity and the accuracy in each season.

D. What this note does and does not do

The note evaluates, over the 11 seasons from 2014/15 to 2024/25 and the 27 states, eight predictors alone and in pairs, a forecasting criterion chosen among them and two baselines, in each of the first 43 weeks of each state's season, reconstructing what would have been available at each date. It also records a forecast for 2026/27, to be checked. It does not forecast the number of cases or the date of the peak, does not go down to the level of municipalities (it only separates the capital from the rest of the state) and does not provide guidance to people or services.

II. DATA

Unit. Each of the 27 states (26 states and the Federal District). Within each state, notifications were split into capital and interior by the municipality of residence reported (IBGE codes [19]); all those without the capital code count as interior, including those without a valid municipality code (see Limitations).

Only probable cases, in all years. The open files for 2014, 2015 and 2022 to 2026 contain only the notifications that were not discarded; those for the other years also contain the discarded ones. So that the series is the same in all years, all counts in this note use only probable cases (notifications not discarded): the outcome, the projection, the growth, the off-season cases and the evidence rule. There are about 24.3 million probable cases from 2007 to 2026. This choice uses the final classification of each notification, which did not exist in real time; anyone following the system also sees notifications that will later be discarded. The effect of this choice was measured in sensitivity S5 (section IV.E) and is large.

One season for each state. The dengue calendar is not the same across the whole country. In the South, Southeast and Center-West, the peak usually comes between February and April; in part of the Northeast, between May and July; in some states in the North, at the end of the year. A single calendar, from July to June, would cut some states' epidemics in half.

Table I. Public data sources used, with period and main limitation.

Source	What it provides	Period	Main limitation
SINAN dengue (OpenDataSUS) [15]	de-identified individual notifications, with date of symptom onset, of notification, of data entry and final classification	files from 2007 to 2026, downloaded on 7 Nov 2026	no data entry date from 2016 to 2020; without the discarded notifications in 2014, 2015 and 2022 to 2026; 2026 file up to the week of 27 Sep 2026
IBGE [18]	population by state and year; population of the municipalities in the 2022 Census	2006 to 2025	2007 and 2023 interpolated
NASA POWER (MERRA-2) [16]	daily mean temperature	2006 to Nov 2026	gridded reanalysis, not station data; 6 points per state; version of 7 Nov 2026
NOAA, ONI index [17]	Pacific temperature anomaly (El Niño)	1950 to Sep 2026	a single number for the whole planet

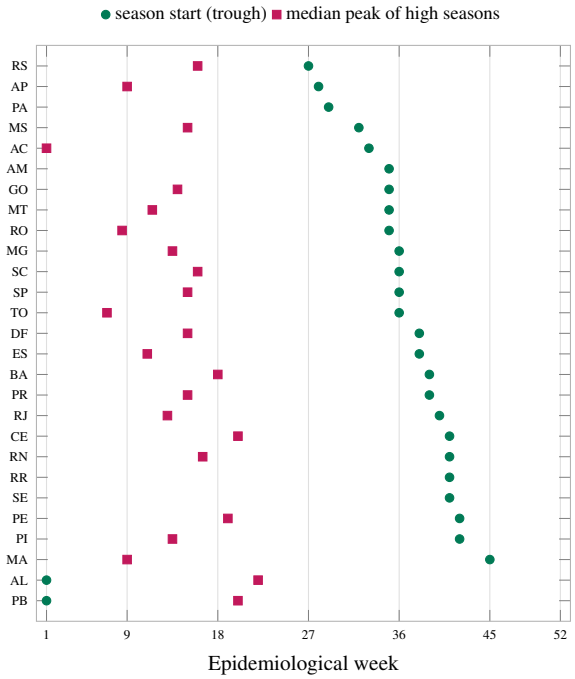


Fig. 1. Epidemiological week of the start of the season (trough) of each state, calculated with the years 2007 to 2013, and median week of the weekly peak of the high-incidence seasons from 2014/15 to 2024/25. States with no high season in the period appear only with the trough.

For that reason each state has its own season, which begins at its **trough week**: the epidemiological week with the lowest mean incidence, calculated with the years 2007 to 2013, prior to the seasons evaluated. The season lasts one year from the trough. In most states, the trough falls between late August and mid-October (weeks 35 to 42); in Alagoas and Paraíba, around the turn of the year; in Rio Grande do Sul, in July (Figure 1).

Outcome. A season is high-incidence when the probable cases with symptom onset in the state's season add up to 300 or more per 100,000 inhabitants, the threshold used in surveillance bulletins for high incidence. A total of 297 state-season pairs were evaluated, of which 136 (45.8%) were high-incidence. Counted from the trough, the median week of the peak of the high seasons was week 28 (half between weeks 23 and 33).

What was known each week. The forecast made on date D should use only what was available on D. Three delays were reconstructed. The first is the **data entry** delay: a case only counts if it was entered before the cutoff. In the files without a data entry date (2016 to 2020) and in the records with an invalid date, the probability that each case was already in the system was estimated from the delay distribution of the same state and the same month in years earlier than or equal to that of the file (2013 to 2015 for the files from 2016 onward). The second is the **publication** delay: the 2026 file, downloaded on 7 Nov 2026, only has complete weeks up to the one beginning on 20 Sep 2026, about 6 weeks earlier; for that reason, the forecast made on date D uses only cases entered up to D minus 6 weeks. The third is the **outcome** delay: the result of a past season is only treated as known 8 weeks after the end of the season plus the publication lag, and this rule applies to the history, to the projection, to the training of the models and to the choice of the criterion. Temperature enters with a 1-week delay and the El Niño index with 2 months. The reconstruction is approximate: the final classification of the cases, the temperature series downloaded today (which may have been revised after the simulated dates [16]) and, in the records without a data entry date, the delay distribution from other years are not exactly what was seen at the time.

III. METHODS

A. Overview of the design

The study simulates what an analyst would have done in each week of the past. For each state, each season from 2014/15 to 2024/25 and each week k of the season, from k = 0 (the trough) to k = 42, the models are trained only on season pairs whose outcome was already known at the date, in the same week k; the forecast is then compared with what happened. Because the states have different calendars, the training for each season and week uses only the outcomes known at the earliest date among the forecasts for that week, which is conservative for the states that start later. That gives 43 weeks $\times$ 11 seasons

Table II. Number of states, out of 27, with a high-incidence season in each season evaluated.

Season	States with high incidence	Season	States with high incidence
2014/15	17	2020/21	10
2015/16	18	2021/22	17
2016/17	3	2022/23	13
2017/18	3	2023/24	19
2018/19	14	2024/25	12
2019/20	10		

× 27 states = 12,771 forecasts per model. The protocol was registered by a dated commit (a9c960e, 7 Nov 2026), before the commit of the results. As in any retrospective analysis of public data, prior registration of the protocol documents the order of the steps, but is not equivalent to a blind study.

B. Two baselines

R0, the history. It is the answer of someone who does not look at the season's data: the fraction of high seasons the state had in the past, among those with a known outcome, with a small (Laplace) correction so as not to give 0% or 100%. For a state that had 13 high seasons out of 19, for example, R0 = (13 + 1)/(19 + 2) = 67%.

A, the count projection. It is the answer of someone who looks at the count and applies a rule of three. It divides the probable cases already available by the fraction that, in the previous seasons of the same state with a known outcome, was already available in the same week relative to the final total. If, in previous years, on average 20% of the season's cases had arrived by this week, and now there are 100 cases per 100,000, the projection is 500 per 100,000:

$$\widehat{I} = \frac{N_{\text{disp}}(k)}{g(k)} \times \frac{100\,000}{\text{pop}} \tag{1}$$

where $N_{\text{disp}}(k)$ is the number of probable cases available in week k and g(k) is the median, over the state's previous seasons (at least 3), of the fraction available in that week. The probability of a high season comes from a simple logistic regression on the logarithm of the projection. In the first weeks, when the lag has not yet let any data from the season arrive, A has no information of its own.

C. Eight candidate predictors

Where do capital and interior come from? The last two predictors were added after the first results (amendment 6). They separate capital and interior because the two parts of the state may behave differently, and each one is compared with the state's own history in the same week of the season. They use counts, and not incidence, because the database does not have the population of each municipality year by year; population growth over 20 years is small compared with the difference between one season and another.

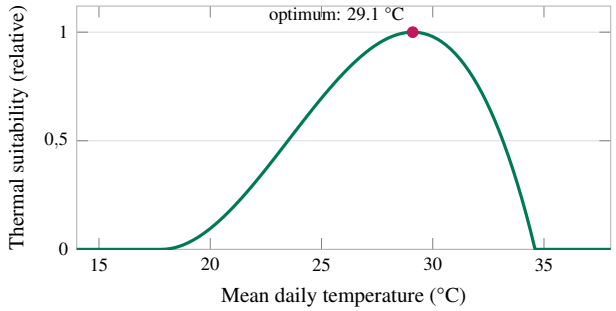


Fig. 2. Thermal suitability curve used in this note: a unimodal approximation built from the limits (17.8 °C and 34.6 °C) and the optimum (29.1 °C) of dengue transmission by *Aedes aegypti* published by Mordecai et al.; it is not the article's mechanistic model.

What is thermal suitability? The *Aedes aegypti* mosquito and the virus depend on temperature in almost everything: how much the mosquito bites, how long it lives, how long it takes to become an adult and how long the virus takes to reach the mosquito's saliva. Mordecai and colleagues combined these traits, measured in the laboratory, into a mechanistic model of transmission potential as a function of temperature, which is zero below 17.8 °C and above 34.6 °C and peaks at 29.1 °C [1]. This note does not reproduce that model: it uses a simple unimodal curve, built for the study, that respects these three values (Figure 2). It was rescaled from 0 to 1 and calculated for each day from the mean temperature of 6 points per state (the capital and the 5 most populous municipalities in the 2022 Census), weighted by population [16, 18, 19]. The original article distinguishes constant temperature from temperature that varies over the day; here, only the daily mean is used. The curve does not measure how many mosquitoes there are, only how much the climate favors transmission.

$$f(T) = \left(\frac{T - 17.8}{11.3}\right)^{\alpha} \left(\frac{34.6 - T}{5.5}\right)^{\beta} \tag{2}$$

for T between 17.8 °C and 34.6 °C, and zero outside that range, with $\alpha + \beta = 3$ and $\alpha = 3 \times 11.3/16.8$, which places the maximum, equal to 1, at 29.1 °C.

Table III. Candidate predictors and the hypothesis behind each one. All measured by state and by forecast date, using only what was available. Capital and interior were added by amendment 6.

Predictor	How it is measured	Hypothesis
Count projection	logarithm of the incidence projected by the rule of three above	what has already arrived, corrected for the state's typical pace, anticipates the total
Recent growth	slope of the logarithm of weekly cases over the last 4 available weeks	a curve that accelerates tends to end high
Off-season cases	logarithm of the incidence with symptom onset in the first 13 weeks of the season (the low season), minus the median of the same value, in the same week, in the state's previous seasons	above-normal transmission when the climate is unfavorable indicates sustained circulation of virus and mosquito, which is amplified when the heat arrives
Mild winter	anomaly of the mean temperature over the state's 91 climatologically coldest days, within the previous 365 days, relative to the 2006 to 2013 mean	a warmer winter leaves the mosquito and transmission uninterrupted
Thermal suitability	mean, over the previous 8 weeks, of a temperature-dependent transmission curve (below) [1]	favorable temperature speeds up the cycle of the mosquito and the virus
El Niño (ONI)	last available value of the ONI index [17]	El Niño alters heat and rainfall in Brazil and has already been used in dengue warnings [3]
Capital	logarithm of the probable cases available with residence in the capital, plus 1, minus the median of the same value, in the same week, in the state's previous seasons	the epidemic usually starts in large cities; the capital above its own normal anticipates the state
Interior	the same for cases with residence outside the capital (in the Federal District, which has no interior, it is zero)	when the interior is already above its own normal, transmission has spread beyond the capital

D. Screening: each predictor alone and in pairs

Each predictor alone (8 models) and each pair (28 models) enters a logistic regression that starts from the state's history and adjusts it with the signals:

$$\operatorname{logit} p = \operatorname{logit} R_0 + \beta_0 + \beta_1 x_1 + \beta_2 x_2 \tag{3}$$

Here, logit is the odds scale (logit p = ln[p/(1 - p)]) and x1, x2 are the predictors standardized in training. A penalization pulls the weights β toward zero, so that the model only moves away from the history when the data give sufficient evidence; its strength is chosen among five values by the lowest error in the last three seasons of each training set. The screening compares the 36 models by the **Brier score** [10], the mean squared error of the probability, summed over all weeks, and by the gain over a baseline, BSS = 1 - Brier(model)/Brier(baseline): positive when the model makes smaller errors. The **incremental gain** of each second predictor is the BSS of the pair with the projection over the projection alone, in the same model: this isolates the effect of the second predictor.

E. Criterion M

Criterion M is the screening model that best forecast the past seasons. To forecast season t, the one among the 36 with the lowest Brier score in the forecasts of the seasons from 2011/12 to t - 2 is chosen, all with an outcome known before any forecast of t. The accuracy of M measures the performance of the whole procedure, including the model changes from one season to another. The composition of M stated in the note is the one chosen with all seasons up to 2024/25. Projection A and criterion M do not have the same form (A does not start from the history and has a fixed penalization); for that reason, the comparison between M and A does not isolate the effect of any predictor, and the gain of each predictor is measured by the incremental gain.

F. How accuracy is measured

Each model gives a probability. When it is 50% or more, the forecast is "high season"; below that, "not high". **Accuracy** is the fraction of state-season pairs in which the forecast matched what happened; **sensitivity** is the fraction of high seasons with a "high" forecast; **specificity**, the fraction of non-high seasons with a "not high" forecast.

Example

In any given week, if the model said "high" for 12 states and "not high" for 15, and got 9 of the 12 and 12 of the 15 right, the season's accuracy is (9 + 12)/27 = 78%. If 11 states had a high season and the model caught 9 of them, the sensitivity is 9/11 = 82%. The week's accuracy is the same calculation summed over the 11 seasons, over 297 pairs.

G. Uncertainty, chance and margin of error

Eleven seasons are few, and the accuracy measured in them varies by chance. To measure this uncertainty, the measures were recalculated 4,000 times over the forecasts already made, drawing with replacement 11 seasons from the 11 available (bootstrap [11]). The models and the choice of M were not refitted in each draw, so these intervals do not include the

uncertainty of fitting or of selection; and the intervals for each week are pointwise, not a simultaneous band for the whole curve. Whole seasons are drawn, not states, because the states are not independent within the same year: when there is a national epidemic, many rise together. Two measures come out of this:

- **95% interval:** the range containing 95% of the resampled values; it indicates, approximately, between which values the true value probably lies.
- **Chance of accuracy of 75% or more:** the fraction of resamples in which accuracy was 75% or more,

$$c = \frac{1}{B}\sum_{b=1}^{B} \mathbf{1}\left[\text{acerto}_b^* \geq 0.75\right], \qquad B = 4\,000 \tag{4}$$

where $\mathbf{1}[\cdot]$ equals 1 when the condition is true and 0 when it is not. The margin of error of the draw (Monte Carlo) is $\pm 1.96\sqrt{c(1-c)/B}$, at most ±1.5 percentage points with B = 4,000. The uncertainty that comes from having only 11 seasons does not decrease with more draws, and it is the one the 95% interval shows. For that reason c is an approximate measure of the stability of the result across draws, and not the probability that the true accuracy exceeds 75%: a point accuracy of 75.4% with an interval of 69.0% to 81.8% shows that the estimate crossed the threshold, not that the true performance is at least 75%. In the screening, the corresponding chance is the fraction of resamples with a positive gain. Following the program's rule, chances of 75% or more are highlighted as promising.

H. Start, evidence and lead time

- **Start (main rule):** the first week k from which accuracy is 75% or more in it and in all following weeks up to k = 42.
- **Start (strict rule):** the same, requiring the lower limit of the 95% interval of accuracy to be 75% or more. Both are reported, because the lead time depends on which one is used.
- **Evident from the count:** a season is evident in week k when the probable cases already available exceed 300 per 100,000. Because only probable cases are counted, evidence implies a high season. At that point no model is needed: counting is enough.
- **Window lead time:** the number of weeks between the start and the first week in which half or more of the high seasons are already evident. If the start comes later, the lead time is zero. The intervals include resamples without a window as zero, and the note reports their fraction.
- **Lead time per season:** for each high season, the week in which the count makes it evident minus the first week from which the model says "high" without interruption up to it; and whether that first week came before the peak. High seasons that do not become evident by week 42 enter with evidence at week 43 (censoring). It includes the states in which the model says "high" from the start based on the history alone, and for that reason it comes with the false alarm rate: the fraction of non-high seasons that the model was calling "high" in the start week.

Why is there no correction for multiple testing? The main question is not a hypothesis test in each week, but the reading of an accuracy curve against a threshold. Looking at 43 weeks, however, increases the chance that one of them exceeds 75% by chance. The protection adopted is the stability rule and the strict rule. In the screening, 36 models were compared; nested selection prevents the luck of any one of them in the evaluated seasons from entering the accuracy of M directly, but it does not erase the decision, made after seeing results, to add capital and interior to the candidates. For this reason, criterion M should also be read as an exploratory result.

I. Sensitivity analyses

Seven variations registered in the protocol: publication lag of 0 weeks (as someone accessing the system without waiting for the open file would have) and of 10 weeks; evidence with 25% and with 75% of the high seasons; exclusion of the atypical seasons from 2015/16 to 2017/18 (Zika and chikungunya circulation) and 2019/20 to 2020/21 (COVID-19 pandemic), also from the calculation of the predictors and the history; and a national season, from July to June, for all states. After external review (amendments 3 and 4), sensitivity S5 was added, which measures the effect of using the final classification of the cases: the count predictors now use all available notifications, an alternative approximation of what was available, while the outcome and the evidence rule continue to use probable cases. Because the files for 2014, 2015 and 2022 to 2026 do not contain the discarded notifications, the clean comparison uses only the seasons whose files contain them (evaluation in 2016/17 to 2020/21), in both versions, identical in everything else.

Protocol deviations. None. Amendment 1 changed only the wording of the protocol; amendment 2 added the additional analysis in section IV.C; amendment 3, made after external review, corrected the denominator of the fraction anticipated before the peak and added sensitivity S5 (final classification) and the aggregate Brier score; amendment 4 defined the clean comparison of S5; amendment 5 corrected the internal validation of the penalization and added the fixed alert rule and the test in held-out seasons (sections IV.F, IV.H and IV.I); amendment 6 added the capital and interior predictors, which enter the screening and the selection of criterion M, and the three-state rule with the projection band (section IV.G), and redid the test in held-out seasons and all sensitivity analyses; amendment 7 made both limits of the band respect the floor of the available incidence and removed that floor in S5 with all notifications. Amendments 2 to 7 were made after the results had been examined, and all are identified as such.

IV. RESULTS

How to read these results

- Weeks are counted from the start of each state's season. For São Paulo, for example, week 0 is early September, week 10 is mid-November, week 18 is mid-January, week 26 is early March and week 35 is mid-May.
- Accuracy is the fraction of state-season pairs with the correct yes-or-no forecast; the threshold is 75%, and the history alone gets 64% to 66% right.
- The 95% interval shows the uncertainty; the chance is the fraction of resamples that exceed the threshold, in bold when it reaches 75%.
- Lead time is the number of weeks between the model starting to be right 75% or more of the time and the count of probable cases making half of the high seasons evident.

A. Screening of the predictors

Alone, three predictors greatly reduced the error of the history: the **interior** (BSS 0.311; 95% CI 0.242 to 0.378), the count projection (0.299; CI 0.246 to 0.352) and off-season cases (0.277; CI 0.185 to 0.371), all with a chance of 100%. The capital alone was well below (0.167; CI 0.099 to 0.242), and thermal suitability, mild winter, El Niño and recent growth, alone, were close to zero. The best pair was **projection and interior** (BSS 0.377 over the history). Added to the projection, in the same model, the interior contributed a BSS of 0.111 (CI 0.062 to 0.161; chance of 100%), off-season cases 0.074 (CI 0.027 to 0.128; chance of 100%) and thermal suitability 0.026 (CI 0.002 to 0.055; chance of 98%); the capital (0.008; chance of 67%), El Niño, mild winter and growth added no clear gain (Table IV). Against projection A, the pair projection and interior had a BSS of 0.074 (CI 0.024 to 0.123; chance of 99.8%).

In the nested selection, M was projection and mild winter in the first season (2014/15), the projection alone in the next three (2015/16 to 2017/18) and **projection and interior** in the last 7 (2018/19 to 2024/25). With all seasons up to 2024/25, the one chosen is **projection and interior**: this is the composition of criterion M from now on. The accuracy of M reported below is that of the procedure, with the model changes over time, and not that of the final pair; the performance of the fixed pair appears in the additional analysis (section IV.C) and, because it was chosen after comparing several candidates, it is a finding to be confirmed.

B. Week-by-week accuracy and lead time

With the 6-week lag of the open file, M exceeded 75% accuracy in **week 17** (75.1%; 95% CI 69.0% to 81.5%; chance of 51%) and stayed above it until the end; under the strict rule, in week 25 (81.5%). Half of the high seasons became evident from the count in **week 35**. The lead time of M was **18 weeks** (95% CI 11 to 29) under the main rule and 10 under the strict rule; the window existed in all resamples. Projection A exceeded 75% in **week 10** (76.1%; CI 71.7% to 80.1%; chance of 70%), with **25 weeks** of lead time (CI 14 to 30); under the strict rule, in week 24, with 11 weeks (Tables V and VI and Figure 3). The median week of the peak of the high seasons was week 28: M opens 11 weeks before it, and A 18 weeks before it.

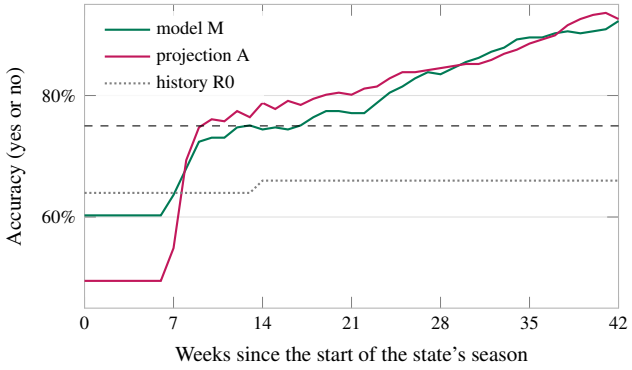


Fig. 3. Accuracy by week of the season with data from the open file (6-week lag): criterion M, projection A and history R0. The dashed line is the 75% threshold.

In accuracy, A was equal to or better than M in 27 of the 35 weeks from week 8 onward. In probability, M was worse than A from weeks 11 to 22 (BSS of -0.035 to -0.002 over A) and better from weeks 23 to 37 (from 0.011 to 0.084). In the aggregate Brier score (lower is better), the history had 0.213, the projection in the model with history 0.150, projection A 0.144, procedure M 0.137, the fixed pair projection and off-season cases 0.138 and the fixed pair projection and interior 0.133. Because A and M do not have the same form, the comparison between them does not isolate the effect of any predictor; the gain of each one is the incremental gain of the screening.

The 75% accuracy is an average. In the start week of M, it identified 78.7% of the high seasons and 72.0% of the non-high ones, and accuracy ranged from 56% to 96% across seasons; in that of A, the sensitivity was 70.6%, that is, almost 3 out of every 10 high seasons still had a "not high" forecast, and accuracy ranged from 63% to 85%.

Per season. Counted season by season, M started saying "high" without interruption, at the median, 24 weeks before the count made the season evident (CI 18 to 28); it anticipated 90% of the high seasons by 4 weeks or more and, in 73.5% of them (CI 65.0% to 82.4%), before the peak. This number is larger than the window's because it includes states in which M says "high" from the beginning based on the history, and it has a cost: in the start week, M was calling 28.0% of the non-high seasons "high". For A, the same numbers were 20 weeks (CI 15 to 22), 90%, 72.8% before the peak (CI 64.7% to 80.1%) and 19.3% false alarm. In these fractions, the high seasons that the model did not anticipate count as not anticipated. In 34 of the 136 high seasons, the count did not reach 300 per 100,000 by week 42, because cases kept arriving afterward; in those, the lead time per season is a minimum.

Table IV. Screening of the 36 models (8 predictors alone and 28 pairs), in order of gain over the history: BSS over R0 and incremental gain over the projection alone in the same model, summed over weeks 0 to 42 of the 11 seasons, with 95% CI and chance of a positive gain (4,000 resamples), and number of seasons in which the model was chosen as M. In bold, chance of 75% or more.

Model	BSS over R0 (95% CI)	Chance	Gain over the projection (95% CI)	Chance	Chosen
projection + interior	0.377 (0.310 to 0.443)	**100%**	0.111 (0.062 to 0.161)	**100%**	7
projection + off-season cases	0.351 (0.272 to 0.431)	**100%**	0.074 (0.027 to 0.128)	**100%**	0
off-season cases + interior	0.330 (0.250 to 0.409)	**100%**	0.044 (-0.027 to 0.117)	**89%**	0
capital + interior	0.320 (0.247 to 0.391)	**100%**	0.030 (-0.039 to 0.105)	**80%**	0
projection + thermal suitability	0.317 (0.263 to 0.374)	**100%**	0.026 (0.002 to 0.055)	**98%**	0
thermal suitability + interior	0.312 (0.236 to 0.387)	**100%**	0.019 (-0.061 to 0.100)	67%	0
interior	0.311 (0.242 to 0.378)	**100%**	0.018 (-0.054 to 0.093)	67%	0
growth + interior	0.305 (0.234 to 0.372)	**100%**	0.009 (-0.062 to 0.083)	58%	0
projection + capital	0.305 (0.236 to 0.378)	**100%**	0.008 (-0.024 to 0.049)	67%	0
mild winter + interior	0.302 (0.233 to 0.376)	**100%**	0.005 (-0.079 to 0.100)	51%	0
El Niño + interior	0.300 (0.222 to 0.381)	**100%**	0.001 (-0.073 to 0.081)	51%	0
projection	0.299 (0.246 to 0.352)	**100%**			3
projection + El Niño	0.297 (0.222 to 0.373)	**100%**	-0.002 (-0.056 to 0.054)	48%	0
projection + growth	0.295 (0.241 to 0.350)	**100%**	-0.006 (-0.015 to 0.005)	14%	0
off-season cases + capital	0.294 (0.212 to 0.377)	**100%**	-0.007 (-0.068 to 0.055)	41%	0
projection + mild winter	0.293 (0.235 to 0.357)	**100%**	-0.008 (-0.063 to 0.040)	40%	1
off-season cases + mild winter	0.281 (0.186 to 0.388)	**100%**	-0.025 (-0.110 to 0.074)	29%	0
off-season cases	0.277 (0.185 to 0.371)	**100%**	-0.031 (-0.106 to 0.045)	21%	0
off-season cases + thermal suitability	0.276 (0.182 to 0.371)	**100%**	-0.033 (-0.113 to 0.051)	21%	0
off-season cases + El Niño	0.270 (0.161 to 0.383)	**100%**	-0.041 (-0.126 to 0.053)	18%	0
growth + off-season cases	0.268 (0.172 to 0.365)	**100%**	-0.044 (-0.125 to 0.039)	15%	0
mild winter + capital	0.178 (0.103 to 0.267)	**100%**	-0.173 (-0.253 to -0.087)	0%	0
thermal suitability + capital	0.175 (0.103 to 0.249)	**100%**	-0.177 (-0.231 to -0.117)	0%	0
capital	0.167 (0.099 to 0.242)	**100%**	-0.188 (-0.232 to -0.138)	0%	0
growth + capital	0.166 (0.096 to 0.241)	**100%**	-0.189 (-0.236 to -0.138)	0%	0
El Niño + capital	0.154 (0.064 to 0.239)	**100%**	-0.207 (-0.270 to -0.147)	0%	0
mild winter + thermal suitability	0.034 (-0.043 to 0.097)	**82%**	-0.378 (-0.519 to -0.247)	0%	0
thermal suitability + El Niño	0.013 (-0.043 to 0.060)	70%	-0.409 (-0.504 to -0.320)	0%	0
growth + mild winter	0.008 (-0.046 to 0.055)	64%	-0.414 (-0.541 to -0.296)	0%	0
growth + thermal suitability	0.008 (-0.034 to 0.051)	64%	-0.414 (-0.549 to -0.301)	0%	0
mild winter	0.004 (-0.055 to 0.056)	57%	-0.421 (-0.554 to -0.296)	0%	0
thermal suitability	0.003 (-0.045 to 0.048)	55%	-0.422 (-0.558 to -0.304)	0%	0
mild winter + El Niño	0.001 (-0.049 to 0.049)	53%	-0.425 (-0.528 to -0.326)	0%	0
growth + El Niño	-0.003 (-0.046 to 0.036)	46%	-0.431 (-0.522 to -0.353)	0%	0
El Niño	-0.008 (-0.055 to 0.034)	39%	-0.437 (-0.536 to -0.347)	0%	0
growth	-0.018 (-0.047 to 0.009)	10%	-0.453 (-0.575 to -0.347)	0%	0

C. Lead time of each screening model (additional analysis)

This analysis was defined after the main results were known (amendment 2 to the protocol) and redone with the 36 models (amendment 6). For each screening model, using the published forecasts, the start, lead time and accuracy were computed with the same rules (Table VII). The fixed pair **projection and interior** opens the window earliest of all models: it passes 75% accuracy at week 9, with **26 weeks** of lead time (95% CI 14 to 29), and, under the strict rule, at week 22, with 13 weeks, the longest strict lead time of all, against 11 for projection A. The interior alone passes 75% at week 14 (21 weeks; 12 under the strict rule). Off-season cases alone pass 75% at week 12 (23 weeks), but this signal is more fragile: under the strict rule it only opens at week 27 (8 weeks), its accuracy does not exceed 81.8%, its pairs with climate signals do not open under the strict rule, and, in the clean comparison of sensitivity S5, with all notifications, it no longer reaches 75% (section IV.E). The capital helps little: alone, it only passes 75% at week 29, and, added to the projection, it delays the window (week 22). Projection A still has the highest maximum accuracy (93.6%). The models with only climate and growth signals never reached 75%.

D. Without the publication lag

Without the 6-week lag, with the data as they are in the notification system on each date, everything happens about 6 weeks earlier, but the lead time barely changes (Figure 4). M exceeds 75% in week 11 (strict rule, 19) and A in week 4 (strict rule, 18); half of the high seasons become evident in week 29. The lead time is 18 weeks for M (CI 11 to 29; 10 under the strict rule) and 25 for A (CI 14 to 30; 11 under the strict rule). The composition chosen for M is still projection and interior, and M starts saying "high" before the peak in 86.0% of the

Table V. Accuracy by week of the season with the 6-week lag of the open file (297 pairs per week): criterion M and projection A, with 95% CI and chance of accuracy of 75% or more, and fraction of the high seasons already evident from the count. In bold, accuracy or chance of 75% or more.

Week	Accuracy M (95% CI)	Chance M	Accuracy A (95% CI)	Chance A	Evident
0	60.3% (54.2% to 65.3%)	0%	49.5% (37.4% to 61.6%)	0%	0%
8	68.0% (59.3% to 75.1%)	3%	69.4% (60.9% to 77.1%)	9%	0%
10	73.1% (66.7% to 79.1%)	28%	**76.1%** (71.7% to 80.1%)	70%	0%
12	74.7% (68.4% to 81.1%)	48%	**77.4%** (72.7% to 82.2%)	**86%**	1%
14	74.4% (68.7% to 80.1%)	43%	**78.8%** (72.7% to 84.8%)	**88%**	1%
16	74.4% (68.4% to 80.8%)	43%	**79.1%** (73.1% to 85.5%)	**90%**	1%
18	**76.4%** (70.7% to 82.5%)	68%	**79.5%** (73.7% to 85.5%)	**94%**	1%
21	**77.1%** (72.1% to 81.8%)	**80%**	**80.1%** (74.7% to 85.5%)	**97%**	4%
24	**80.5%** (74.1% to 86.5%)	**96%**	**82.8%** (77.4% to 88.2%)	**100%**	8%
26	**82.8%** (77.8% to 87.9%)	**100%**	**83.8%** (78.8% to 88.9%)	**100%**	11%
28	**83.5%** (78.5% to 88.2%)	**100%**	**84.5%** (79.5% to 89.6%)	**100%**	20%
30	**85.5%** (81.5% to 89.9%)	**100%**	**85.2%** (80.8% to 89.9%)	**100%**	26%
35	**89.6%** (86.9% to 92.6%)	**100%**	**88.6%** (85.5% to 91.6%)	**100%**	50%
38	**90.6%** (87.5% to 93.3%)	**100%**	**91.6%** (89.2% to 93.9%)	**100%**	61%
42	**92.3%** (89.9% to 94.6%)	**100%**	**92.6%** (90.2% to 94.9%)	**100%**	75%

Table VI. Performance in the start week, under the main and strict rules: overall accuracy, sensitivity (high seasons identified), specificity (non-high seasons identified) and range of accuracy across the 11 seasons.

Criterion and rule	Week	Accuracy	Sensitivity	Specificity	Accuracy per season
M, main rule	17	75.1%	78.7%	72.0%	56% to 96%
M, strict rule	25	81.5%	80.1%	82.6%	67% to 96%
A, main rule	10	76.1%	70.6%	80.7%	63% to 85%
A, strict rule	24	82.8%	76.5%	88.2%	70% to 96%

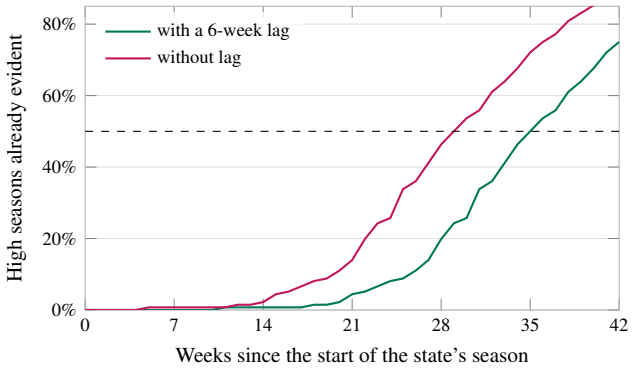


Fig. 4. Fraction of the high seasons already evident from the count (probable cases available above 300 per 100,000) in each week of the season, with the 6-week lag of the open file and without lag. The dashed line marks half.

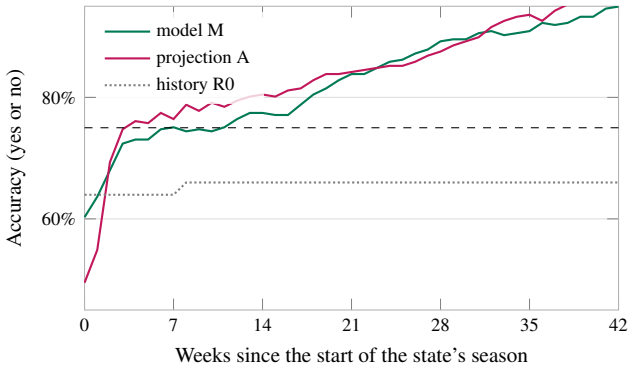


Fig. 5. Accuracy by week of the season without publication lag: criterion M, projection A and history R0. The dashed line is the 75% threshold. The comparison with Figure 3 shows the effect of the 6-week lag of the open file.

high seasons.

E. Sensitivity analyses

The final composition of M was projection and interior in all analyses except the national season, in which it was projection and off-season cases. The lead time of M stayed at 18 weeks with 0, 6 and 10 weeks of lag, and that of A at 25; what changes with the lag is the week in which everything happens. Without the atypical seasons, also excluded from the calculation of the predictors and the history, M only exceeds 75% in week 26, and the lead time drops to 10 weeks, with an interval of 3 to 25 and 0.4% of the resamples without a window; A keeps 23. The measure depends on how much is required of the count: with 25% of the high seasons evident, M anticipates by 13 weeks; with 75%, by 25. With the national season, from July to June, half of the high seasons do not become evident by week 42 of the season, and the lead times become minimums (23 weeks or

Table VII. Additional analysis (amendment 2): start under the main rule, lead time relative to week 35 (half of the high seasons evident) with 95% CI, fraction of resamples without a window, start and lead time under the strict rule, and maximum accuracy, for the 26 screening models that reached 75% accuracy and the two references, in order of start. The other 10 screening models (only with climate and growth signals) did not reach 75% in any week. In bold, the note's two references.

Model	Start	Lead time (95% CI)	No window	Strict rule	Maximum accuracy
projection + interior	9	26 (14 to 29)	0.0%	22 (13)	92.9%
projection A	10	25 (14 to 30)	0.0%	24 (11)	93.6%
El Niño + interior	10	25 (11 to 29)	0.0%	26 (9)	87.9%
off-season cases + capital	10	25 (5 to 29)	0.1%	32 (3)	87.2%
off-season cases + El Niño	11	24 (0 to 29)	9.9%	no window	83.5%
growth + off-season cases	11	24 (0 to 29)	8.2%	no window	81.8%
off-season cases	12	23 (8 to 29)	2.3%	27 (8)	81.8%
capital + interior	14	21 (11 to 28)	0.0%	26 (9)	89.9%
off-season cases + interior	14	21 (13 to 29)	0.0%	24 (11)	88.9%
interior	14	21 (13 to 27)	0.0%	23 (12)	88.2%
thermal suitability + interior	14	21 (12 to 28)	0.0%	24 (11)	88.2%
mild winter + interior	14	21 (9 to 29)	0.0%	26 (9)	87.2%
off-season cases + thermal suitability	14	21 (0 to 29)	8.4%	no window	81.1%
projection + off-season cases	16	19 (11 to 29)	0.0%	25 (10)	92.6%
growth + interior	16	19 (12 to 28)	0.0%	25 (10)	87.5%
off-season cases + mild winter	16	19 (0 to 29)	7.5%	no window	81.5%
criterion M	17	18 (11 to 29)	0.0%	25 (10)	92.3%
projection + thermal suitability	18	17 (11 to 24)	0.0%	25 (10)	92.9%
projection	19	16 (10 to 23)	0.0%	26 (9)	91.6%
projection + growth	19	16 (10 to 23)	0.0%	26 (9)	90.9%
projection + mild winter	20	15 (9 to 23)	0.0%	25 (10)	91.6%
projection + El Niño	20	15 (10 to 22)	0.0%	25 (10)	90.9%
projection + capital	22	13 (10 to 25)	0.0%	26 (9)	93.3%
capital	29	6 (1 to 16)	1.1%	no window	86.5%
growth + capital	29	6 (0 to 15)	2.9%	no window	85.2%
thermal suitability + capital	32	3 (2 to 16)	0.2%	34 (1)	86.2%
mild winter + capital	32	3 (0 to 18)	3.9%	no window	85.5%
El Niño + capital	32	3 (1 to 17)	2.0%	no window	84.8%

Table VIII. Composition of M, start under the main rule, evidence week (half of the high seasons evident, or 25% and 75% in the last two rows) and lead time with 95% CI (resamples without a window counted as zero), for criterion M and projection A. Weeks of the season; "after 42" indicates that evidence was not reached by the last week evaluated, and the lead time is then a minimum. The S5 rows measure the effect of the final classification: the mixed version uses all notifications where the file contains them; the two clean versions evaluate only the 2016/17 to 2020/21 seasons, with evidence at week 40.

Analysis	Composition of M	Start M	Start A	Evidence	Lead time M	Lead time A
Main (6-week lag)	projection + interior	17	10	35	18 (11 to 29)	25 (14 to 30)
No lag	projection + interior	11	4	29	18 (11 to 29)	25 (14 to 30)
10-week lag	projection + interior	21	14	39	18 (11 to 29)	25 (14 to 30)
Without atypical seasons	projection + interior	26	13	36	10 (3 to 25)	23 (6 to 27)
National season (July to June)	projection + off-season cases	20	9	after 42	$\geq$ 23 (6 to 35)	$\geq$ 34 (4 to 35)
Evidence with 25% of high seasons	projection + interior	17	10	30	13 (5 to 22)	20 (8 to 23)
Evidence with 75% of high seasons	projection + interior	17	10	42	25 (17 to 34)	32 (19 to 35)
Mixed S5, all notifications	projection + interior	28	23	35	7 (3 to 19)	12 (5 to 22)
Clean S5, probable cases	projection + interior	24	14	40	16 (10 to 21)	26 (15 to 33)
Clean S5, all notifications	projection + interior	29	22	40	11 (1 to 15)	18 (10 to 29)

more for M and 34 or more for A). In all versions, the interior added to the projection reduced the error in the same model (incremental BSS of 0.06 to 0.14; chance of 99% or more), more than off-season cases in most analyses and with a very close value in clean S5 with probable cases only. In sensitivity S5, the effect of the final classification appears clearly. In the mixed version, with all notifications where the file contains them, the lead time of M drops to 7 weeks (2 under the strict rule) and that of A to 12 (4 under the strict rule), but this version mixes two types of count. In the clean comparison, in the 2016/17 to 2020/21 seasons, the lead time went from 16 to 11 weeks for criterion M (from 13 to 5 under the strict rule), from 26 to 18 for projection A (from 18 to 12), from 18 to 15 for the pair projection and interior (from 15 to 11) and from 20

to 13 for the pair projection and off-season cases (from 13 to 8); off-season cases alone, which had 5 weeks, no longer reached 75%. Both pairs continued to improve the projection in the same model in both versions (incremental BSS of 0.061 and 0.104 with the interior and of 0.062 and 0.065 with off-season cases; chance of 100%). With only 5 seasons, these intervals are wide. In this version, 95 of the 135 seasons were not high: always answering "not high" would already be right 70.4% of the time, which helps to read the 75% threshold.

F. Fixed alert rule (additional analysis)

The measures above come from aggregate curves and from a stability rule that can only be verified looking back. To measure what a user would actually have had, an alert rule fixed before the calculation (amendment 5) was applied, the same for all models: the alert is the first week in which the probability is 50% or more in it and in the previous week, and only the first alert counts. For each high season, the number of weeks by which the alert came before the count of probable cases exceeded 300 per 100,000 is measured (zero if it did not come before), as well as whether it came before the peak; for each non-high season, whether there was a false alarm at any point up to week 42. The rule was defined after the earlier results had been examined; it is an additional retrospective analysis, not a validation (Table IX). The pair projection and interior was added to the table with amendment 6.

The rule shows the cost that the aggregate curves hide. **The history alone already gives an alert** before the count in 71% of the high seasons, because in the states that usually have a high season it says "high" from the start; and it raises a false alarm in 37% of the non-high seasons. Projection A gives an alert before evidence in 97% of the high seasons and before the peak in 86%, but with a false alarm in 42%; M and the two pairs are similar, with 46% to 48% false alarms. With the count less dependent on the final classification (clean S5, seasons 2016/17 to 2020/21), almost all of them get worse: projection A drops to 75% of the high seasons with an alert and 57% false alarms; the pair projection and interior ends up with 85% and 40%, and the pair projection and off-season cases with 78% and 39%, the only two that, in this version, give an alert in more high seasons than the history (62%) with fewer false alarms than it (42%). In all versions, about 4 out of every 10 non-high seasons would receive a false alarm with this rule.

G. Three-state rule with the projection band (additional analysis)

Amendment 6 tested a rule that only declares the result when the whole probable band of the count projection lies on one side of the threshold. In each week, the band comes from the errors the projection had, in the same week of the season, in all past state-season pairs with an outcome already known (10% and 90% quantiles, that is, an 80% band). The two limits never fall below the incidence already available; in clean S5 with all notifications, this floor is not used, because the available incidence of probable cases depends on the final classification. The forecast is in the **high** state when the lower limit of the band exceeds 300 per 100,000, in the **low** state when the upper limit is below 300, and in the **undetermined** state in between. The alert follows the rule of the previous section: high state in that week and in the previous one, first alert only. The 90% band was the only sensitivity analysis registered (Table X).

The band does what it promises with false alarms and charges dearly in lead time. In the main analysis, false alarms drop to 2% (3 of the 161 non-high seasons; CI 0% to 4%), against 42% for projection A with the probability rule. The accuracy of 96% applies only to the forecasts in which the rule decides; 58% remain undetermined. The rule also errs in the other direction: in some week, it declared low 21 of the 136 seasons that ended high (15%); in clean S5 with all notifications, 10 of 40 (25%). In exchange for the few false alarms, only 75% of the high seasons receive an alert before evidence, and the alert comes, at the median, only 4 weeks before the count (CI 2 to 10), before the peak in 40% of them. The 90% band is even slower. With the count less dependent on the final classification (clean S5, all), the rule, which in this version is also computed without the floor of the available incidence, almost stops anticipating: 25% of the high seasons with an alert before evidence and a median lead time of zero; no false alarm was observed in the 95 non-high seasons, but, with no events, resampling cannot measure the uncertainty of this rate, and the true rate may not be zero. The 80% band covered 78% of the outcomes in the main analysis, close to the nominal value, but only 73% in clean S5 with all notifications. In the data evaluated, the projection band is too wide early in the season to decide early, and a "low" state does not rule out a high season.

H. Test in held-out seasons (additional analysis)

To test the model choice on data it had not seen (amendment 5), the screening model with the lowest Brier score in the seasons from 2011/12 to 2018/19 only was chosen, and that model, projection A and the history were evaluated only in 2020/21 to 2024/25. With the 21 models of amendment 5, the choice fell on projection and El Niño, which had a Brier score of 0.160 in the held-out seasons, worse than that of projection A (0.147): this test confirmed the count projection, and not the advantage of any second predictor. Redone with the 36 models (amendment 6), the choice fell on **projection and interior**, which had a Brier score of 0.129 in the held-out seasons, lower than that of projection A (0.147); it anticipated the count by 27 weeks under the main rule and 9 under the strict rule, against 26 and 5 for projection A; and, with the alert rule, it gave an alert before evidence in 99% of the high seasons, with 42% false alarms, against 96% and 27% for projection A and 68% and 30% for the history. These seasons, however, had already been used in the test of amendment 5, and the idea of separating capital and interior arose after results that included them had been examined; the interior predictor had not been calculated before amendment 6 was registered, but the result counts as a description, and not as a clean test.

Table IX. Fixed alert rule (two consecutive weeks with probability of 50% or more; first alert only), applied from week 0 to week 42 of each state's season: median lead time of the alert relative to evidence from the count in the high seasons (zero when there was no alert before), with 95% CI; fraction of the high seasons with an alert before evidence and before the peak; and fraction of the non-high seasons with a false alarm, with 95% CI. In the main analysis, 34 of the 136 high seasons do not become evident by week 42 and enter with evidence at week 43; for them, the lead time is truncated and is a minimum value.

Analysis and model	Median lead time (95% CI)	Alert before evidence	Alert before peak	False alarm (95% CI)
Main: history R0	27.5 (26 to 32)	71%	71%	37% (30% to 44%)
Main: projection A	22.5 (20 to 30)	97%	86%	42% (23% to 65%)
Main: criterion M	29.5 (27 to 33)	97%	85%	48% (40% to 57%)
Main: projection + off-season	28 (27 to 33)	96%	84%	46% (36% to 55%)
Main: projection + interior	28.5 (27 to 33)	98%	85%	48% (38% to 57%)
Clean S5, all: history R0	30 (25 to 35.5)	62%	62%	42% (38% to 45%)
Clean S5, all: projection A	19 (7 to 34)	75%	68%	57% (10% to 88%)
Clean S5, all: criterion M	28 (23 to 34)	85%	70%	49% (33% to 64%)
Clean S5, all: projection + off-season	28.5 (22 to 34)	78%	72%	39% (28% to 47%)
Clean S5, all: projection + interior	28 (23 to 34)	85%	70%	40% (33% to 45%)

Table X. Three-state rule with the projection band (amendment 6), with an 80% or 90% band and an alert in two consecutive weeks in the high state: median lead time of the alert relative to evidence from the count in the high seasons (zero when there was no alert before), with 95% CI; fraction of the high seasons with an alert before evidence and before the peak; false alarm in the non-high seasons, with 95% CI; fraction of the high seasons declared low in some week (not necessarily in two consecutive ones); fraction of the forecasts (state, season and week) in the undetermined state; and accuracy among the decided states (high or low). Clean S5 with the 80% band.

Analysis and band	Lead	Before evidence	Before peak	False alarm	Highs called low	Undetermined	Accuracy
Main, 80%	4 (2 to 10)	75%	40%	2% (0% to 4%)	15%	58%	96%
Main, 90%	2 (0 to 5)	60%	35%	1% (0% to 3%)	10%	66%	97%
Clean S5, probable	1 (0 to 5)	62%	28%	2% (0% to 4%)	22%	53%	96%
Clean S5, all	0 (0 to 0)	25%	20%	none (0 of 95)	25%	60%	95%

I. Correction of the internal validation

In the choice of the penalization, the internal validation now uses only outcomes known at the date, as the external training already did (amendment 5). The chosen penalization changed in 14.7% of the 12,348 fits; the probabilities changed, on average, by 0.002 in the projection and in the pair and by 0.006 in criterion M; with criterion M still chosen among the 21 models of amendment 5, the Brier score of M went from 0.148 to 0.147, and the lead time of M remained at 17 weeks under the main rule and went from 9 to 10 under the strict rule. Projection A does not use this step and did not change. All numbers in this note are already those of the corrected code.

J. Registered forecast for 2026/27

With the files downloaded on 7 Nov 2026 and the seasons with a known outcome on that date, the models produced the forecast in Table XI, dated 1 Nov 2026 (data up to 20 Sep 2026), for the 24 states whose season had already begun. In Alagoas, Paraíba and Maranhão, the season had not yet begun. The forecast was recorded by commit before any later data were downloaded and will be checked when there is a file covering the whole season. Column M was redone with the composition of amendment 6, still with the same files, and the Band column gives the state under the three-state rule; R0 and A did not change. In these early weeks (2 to 17), the historical accuracy of M lies between 60% and 75%; the table is a record for checking, and not a reliable forecast.

M gives 50% or more in 14 of the 24 states, against 10 for A and 13 for R0. The projection band already classifies 7 states as high (GO, AC, MS, MT, TO, MG and SC) and none as low; in Santa Catarina, where the projection is very high and the history is low, M stays at 46%. In several states, the season was in weeks 2 to 6 and the lag had not yet let cases from the season arrive; there, A has no information of its own and M stays close to the history.

V. DISCUSSION

A. Main findings

In this retrospective simulation, the most robust result is still the case count projection: it exceeds 75% average accuracy in week 10 of the season, 25 weeks before the count makes half of the high seasons evident (11 under the strict rule), and 18 weeks (12) with the count less dependent on the final classification. The best complement found was the **interior**, that is, the cases with residence outside the capital compared with the state's own history: added to the projection, it reduced the error in the same model in all versions (incremental BSS of 0.06 to 0.14), more than off-season cases in most of them (in clean S5 with probable cases only, the values were very close: 0.061 and 0.062, which does not demonstrate equiva-

Table XI. Probability of a high-incidence season in 2026/27, by state, on 1 Nov 2026: week of the season on that date, history R0, projection A, criterion M (projection and interior) and state from the projection band. A record for checking, not a warning.

State	Week	R0	A	M	Band
GO	9	95%	94%	97%	high
AC	11	75%	82%	81%	high
MS	12	70%	84%	79%	high
SP	8	65%	65%	75%	undetermined
MT	9	65%	81%	73%	high
ES	6	70%	46%	70%	undetermined
TO	8	55%	77%	69%	high
MG	8	55%	83%	69%	high
DF	6	65%	46%	65%	undetermined
PR	5	60%	46%	60%	undetermined
RN	3	60%	46%	60%	undetermined
RO	9	45%	75%	57%	undetermined
CE	3	50%	46%	50%	undetermined
BA	5	50%	46%	50%	undetermined
SC	8	25%	96%	46%	high
RJ	4	40%	46%	40%	undetermined
RS	17	24%	24%	36%	undetermined
PE	2	35%	46%	35%	undetermined
AP	16	33%	24%	32%	undetermined
RR	3	30%	46%	30%	undetermined
PI	2	25%	46%	25%	undetermined
AM	9	15%	50%	18%	undetermined
PA	15	10%	46%	12%	undetermined
SE	3	10%	46%	10%	undetermined

lence); the fixed pair projection and interior had the lowest Brier score and the longest lead time under the strict rule (13 weeks), and was the one chosen when the selection used only the seasons from 2011/12 to 2018/19. The capital, contrary to the hypothesis, added no clear gain to the projection, although alone it improves on the history. Criterion M, which became projection and interior, opens the window later than projection A under the main rule (18 against 25 weeks), because the procedure changed models over time. With a probability alert rule, about 4 out of every 10 non-high seasons receive a false alarm; requiring the whole projection band to exceed the threshold practically eliminates false alarms, but also almost all of the lead time.

B. Promising aspects and limits, side by side

On the promising side, with a chance of 75% or more:

- the window of projection A existed in all resamples, in the main analysis and with the count less dependent on the final classification;
- the gain of the interior added to the projection, in the same model, in all versions (chance of 99% or more), and that of off-season cases;
- the accuracy of M of 78.8% or more from week 23 and that of A of 80% or more from week 19 (chances above 85%);
- the three-state rule, with a false alarm rate of 2% (CI 0% to 4%) and accuracy of 96% in the forecasts in which it decides, which are 42% of the total.

On the negative side, at the start under the main rule the accuracy of M has an interval that goes down to 69%, and the chance of exceeding 75% in that week is 51%; under the strict rule, the lead time drops to 10 weeks for M and 11 for A. Average accuracy of 75% is not 75% in each season: there were seasons with 56% (M) and 63% (A) at the start. With the count less dependent on the final classification, the lead time drops by 3 to 8 weeks, M is left with 5 weeks under the strict rule and off-season cases alone no longer work. The advantage of the pair projection and interior in the held-out seasons does not count as a clean test. With the probability alert rule, false alarms range from 37% to 57% of the non-high seasons; with the three-state rule, they drop to 0% to 2%, but the median lead time drops to 0 to 4 weeks, and 10% to 25% of the high seasons are declared low in some week.

These highlights come from curves read over 43 weeks, from 36 models and from several analyses; some of them could appear by chance. For that reason the note only counts as the start the point at which accuracy stays stably above 75%, reports the strict rule, and isolated numbers should be read as a signal, not as proof.

C. Comparison with the literature

The pattern is consistent with what other studies have found. In the open challenge in the Americas, the models with more biology were not better than the simple ones [5]; here, no climate signal alone improved on the history, and the best complements to the projection were other signals from the count itself: the interior and off-season cases. The capital above its own normal did not anticipate the state, but the interior did; this fits the idea that a large season is one that spreads beyond the capital, but the note does not test this mechanism. In Vietnam, the gain over the historical baseline existed at 1 to 3 months ahead, but not at 4 to 6 [6]; here, the lead time is measured against a detection rule based on cumulative incidence, and not against the history, with another horizon and another target; the numbers are not directly comparable. The importance of the reporting delay is what motivates InfoDengue's nowcasting [2] and the studies with digital data [9].

D. Strengths

- Lead time measured against the count of probable cases itself, with two start rules and with the fraction of seasons anticipated before the peak.
- Season defined by state and a series of probable cases only, the same in all years.
- Training, history and criterion selection cut off by the date on which each outcome became known.
- Screening of eight predictors alone and in pairs, with the incremental gain of each one measured in the same model.
- Two fixed alert rules, one by probability and another by the projection band, which show the trade-off between lead time and false alarm.
- All 27 states, 11 seasons and 43 weeks, with uncertainty that respects the dependence between states in the same year; registered prospective forecast; individual forecasts, code and aggregated data open.

E. Weaknesses of the method

- **Partial reconstruction of the past.** The final classification of the cases, the current climate series and the delay distribution from other years are not exactly what was seen at each date. The effect of the final classification was measured and is large: with the count less dependent on it, the lead time dropped by 3 to 8 weeks in the seasons in which it could be measured. The alternative version does not reproduce the system of the time either: it includes notifications that could already have been discarded and keeps the evidence rule based on the final classification. The numbers of the main analysis are therefore a potentially optimistic estimate.
- **Low statistical power.** With 11 seasons, the interval of the lead time of M runs from 11 to 29 weeks.
- **Yes-or-no outcome.** The fixed threshold of 300 per 100,000 treats a season of 310 and one of 1,000 the same way.
- **Unit too large.** Epidemics start in cities; the state average dilutes local signals. Separating capital and interior is only a first step.
- **Where the gain of the interior comes from.** The interior is compared with the state's own history. The note did not test the state total compared with itself, and therefore does not know whether the gain comes from the separation between capital and interior or from that comparison.
- **Trough estimated with 7 years.** The start week of each state was calculated with 2007 to 2013 and may have changed.
- **Conservative training.** The training for each week uses the outcomes known at the date of the state that starts earliest, which removes information from the others.
- **Constructed lead time.** The main measure compares two aggregate curves; it is not the lead time in a specific state, nor relative to the start of the epidemic, the peak or the need to expand care. It grows when the count milestone comes later: 13, 18 and 25 weeks for M with 25%, 50% and 75% of the high seasons evident.
- **Stability seen afterward.** The rule that accuracy stays above 75% up to week 42 can only be verified looking back; an analyst in week 17 would not know it. For future use, the week in which the warning is issued and the evaluation criteria need to be fixed before the season.
- **Model M anchored in the history.** M tends to say "high" early in the states that usually have a high season, which increases the lead time per season and also the false alarm rate.
- **Notification is not infection.** Changes in care seeking, testing and notification, including with the arrival of the vaccine, alter the outcome without changing transmission.

F. What the result allows us to conclude

It allows us to conclude that, in this retrospective simulation, the case count projection anticipates, with average accuracy of 75%, which states will have a high season before the count of probable cases makes it evident, also when the count is less dependent on the final classification and in the held-out seasons; and that the interior adds information to the projection within the same model, in general more than off-season cases. It does not allow the weeks of lead time to be treated as demonstrated operational capacity: they are aggregate measures, they

depend on the start rule, on the count milestone and on the final classification; a simple alert rule raises many false alarms, and requiring the whole band above the threshold eliminates almost all of the lead time and does not prevent high seasons from being declared low. It does not demonstrate an advantage over surveillance alerts, which were not compared. It does not allow off-season cases alone to be used as a warning signal, nor the claim that the capital, climate or El Niño do not matter for dengue.

G. Next steps

Each path below only becomes work with a new protocol: check the forecast registered for 2026/27, including the states given by the projection band; test the state total compared with its own history, to find out where the gain of the interior comes from; redo the analysis with the InfoDengue weekly series [2], through the Mosqlimate API [20], which includes notifications not yet classified; test off-season cases with other definitions of the low season; and go down to health regions or large municipalities.

VI. CONCLUSION

In this retrospective simulation, the case count projection is the result that survived all checks: average accuracy of 75% or more from week 10 of the state's season, 25 weeks before the count makes half of the high seasons evident (11 under the strict rule), and 18 weeks (12) with the count less dependent on the final classification. In the main analysis, which uses the final classification of cases, with criterion M, whose final composition is the case count projection and the cases in the state's interior (in the retrospective forecasts, M changed models over the seasons), a high season can be forecast 18 weeks ahead of the simple case count (95% CI 11 to 29 weeks; 10 under the strict rule); with the count less dependent on the final classification, 11 weeks in seasons 2016/17 to 2020/21 (5 under the strict rule). Criterion M did not open the window before the projection alone. The interior improved the projection in all versions, more than off-season cases in most of them and with a very close value in clean S5 with probable cases only, and the fixed pair projection and interior had the lowest error and the longest lead time under the strict rule (13 weeks), but it was chosen after comparing 36 models and still needs confirmation; the capital added no clear gain to the projection. With a probability alert rule, about 4 out of every 10 non-high seasons would receive a false alarm; requiring the whole projection band above the threshold eliminates almost all false alarms, but also almost all of the lead time, and still declares low, in some week, part of the seasons that end high. These numbers measure the lead time relative to the count, in aggregate form, and not the warning each state would receive before the start of the epidemic; checking the forecast registered for 2026/27 is the next test.

VII. LIMITATIONS

- The research steps (collection, database assembly, calculation of the predictors, models, analysis and writing) were carried out with artificial intelligence and did not undergo independent human validation. The results may contain errors or gaps because of this.
- The design weaknesses listed in the Discussion also apply as limitations.
- As in any retrospective analysis of public data, prior registration of the protocol documents the order of the steps, but is not equivalent to a blind study; checking the forecast registered for 2026/27 is the prospective test.
- The separation between capital and interior uses the municipality of residence reported in the notification. Notifications with a municipality code that is missing, has fewer than 6 digits or belongs to another state count as interior; the codes were not checked against the official list of municipalities. This group makes up 0.01% of probable cases and at most 4.7% in one state and year (Santa Catarina, 2014: 6 of 129 cases). The low frequency reduces the concern that it explains the interior signal, but the effect was not measured, because the analysis was not redone without these records.
- Dengue vaccination and circulating serotypes were not included in the models.
- The forecast for 2026/27 does not cover Alagoas, Paraíba and Maranhão.

VIII. GLOSSARY

Technical terms used in this note, with attention to those that are not common in traditional epidemiology studies.

- **95% confidence interval:** range of values compatible with the data.
- **Accuracy:** fraction of state-season pairs in which the yes-or-no forecast matched what happened.
- **Bootstrap:** redoing the calculation many times by drawing the data with replacement, to measure how much the result varies by chance.
- **Brier and BSS:** mean squared error of the probability; the BSS is the fraction of that error that one model eliminates relative to another.
- **Capital and interior:** cases with residence in the state capital and in the other municipalities; here, each part compared with its own history in the same week of the season.
- **Censoring:** when the expected event does not happen by the end of follow-up; the measured value is then a minimum.

- **Chance of accuracy of 75% or more:** fraction of resamples in which accuracy was 75% or more; from 75% upward, the note calls it promising.
- **Criterion M:** the screening model that best forecast the past seasons, chosen without looking at the season being forecast.
- **El Niño and ONI:** periodic warming of the Pacific Ocean that alters the climate in Brazil; the ONI is the index that measures this warming.
- **Epidemiological week:** Sunday-to-Saturday week used by surveillance.
- **Evident from the count:** a season whose probable cases already available have exceeded 300 per 100,000.
- **Expanding window:** scheme in which each season is forecast using only the previous ones.
- **False alarm:** a season that was not high and that the model was calling "high".
- **Imputation:** estimating a missing value from similar data; here, the data entry date from 2016 to 2020.
- **Incremental gain:** how much one predictor improves another within the same model.
- **Known outcome:** result of a past season treated as available; here, 8 weeks after the end of the season plus the publication lag.
- **Lead time:** number of weeks between the model starting to be right 75% or more of the time and the count making half of the high seasons evident.
- **Logistic regression:** model that turns signals into a probability between 0 and 1.
- **Mild winter:** a winter warmer than the state's average, measured over the 91 normally coldest days.
- **Monte Carlo error:** imprecision of a number calculated by random draws; it decreases with more draws.
- **Nested selection:** choosing the model using only the seasons before the one to be forecast.
- **Nowcasting:** estimating the current number of cases by correcting for those not yet reported.
- **Off-season cases:** incidence in the state's low season above or below the normal of previous seasons.
- **Offset:** fixed term of a model; here, the state's history, from which the predictors can only move the forecast away.
- **Penalization:** rule that shrinks a model's weights to keep it from learning noise.
- **Probable case:** dengue notification that was not discarded during investigation.
- **Projection band:** interval within which, based on the errors of past seasons in the same week, the season total is expected to fall; here, 80%.
- **Publication lag:** time between a case entering the notification system and appearing in the open file; here, about 6 weeks.
- **Reanalysis:** gridded reconstruction of past climate, combining observations and physical models; here, MERRA-2.
- **Screening:** comparison of each predictor alone and in pairs.
- **Sensitivity:** fraction of the high seasons that the model called "high".
- **Specificity:** fraction of the non-high seasons that the model called "not high".
- **Strict rule:** start requiring the lower limit of the 95% interval of accuracy to exceed 75%.
- **Thermal suitability:** number from 0 to 1 that indicates how much temperature favors dengue transmission by *Aedes aegypti*; here, an approximation that is zero in extreme cold and heat and peaks at 29.1 °C.
- **Three-state rule:** rule that only says "high" or "low" when the whole projection band lies on one side of the threshold, and "undetermined" otherwise.
- **Trough week:** epidemiological week with the lowest mean dengue incidence in a state; here, the start of that state's season.

IX. REPORTING GUIDELINES

The note follows, insofar as they apply to population-level forecasting, the TRIPOD+AI [12], EPIFORGE [13] and RECORD [14] guidelines. The main items are in the sections: context and questions (I), sources, period, probable case series, season by state, outcome and information available at each date (II), design, predictors, screening, criterion M, measures, uncertainty and criteria (III), screening, performance with uncertainty, sensitivity and specificity, lead time, sensitivity analyses and alert rules (IV.A to IV.I), registered prospective forecast (IV.J), comparison with the literature and weaknesses (V), limitations (VII), use of AI, data and code (form at the end). The individual forecasts by state, season and week are published. There are no individual data on people; all tables are by state.

DISCLAIMER

This is a working note in health informatics, produced with extensive use of artificial intelligence, and not a clinical study. The research steps carried out with AI (data collection, literature screening, analysis and code) have not undergone independent human validation, and the results may contain errors or gaps as a consequence; the author reviews the final text before publication. It has not been reviewed by a licensed health professional. It does not constitute advice, diagnosis, prognosis or treatment recommendation for any person. Hypotheses are hypotheses, not confirmed findings. The source data have limitations described in the text. Any use in health care requires specific review by licensed professionals, according to the intended use. Valid for the date and version indicated.

ARTIFICIAL INTELLIGENCE USE STATEMENT

Tools: Claude Opus 5.5 (Anthropic), in sessions on 7 and 8 Nov 2026. Steps with AI: literature search, protocol design, all code (collection, database, signals, models, evaluation, figures), running the analyses, writing and translation into English and Spanish. Checking: automatic only: the code runs end to end from raw files with SHA-256 hashes; the numbers in the text were checked by script against the result files; DOIs and links were verified. There was no independent human validation of the AI steps, and the results may contain errors or gaps because of this; the author reviewed the final text. Tool limitations: it can make mistakes in code and in reading the literature that automatic checks do not catch. Record: protocol, code and results versioned in the program's public repository (github.com/willianwistuba/mais-saude-ai), with the protocol in commit a9c960e, the 2026/27 forecast in commit dba78f5, amendment 6 in commit 3149434, its results in commit 252e1a5 and amendment 7, with its results, in commit c054974. Conflicts and funding: none.

DATA AND CODE AVAILABILITY

All data used are public, and the scripts that produce every number are available at https://mais-saude-ai.vercel.app/en/code/, with instructions to reproduce the results.

ETHICS

This note uses only public, aggregated data that do not identify individuals, which waives ethics committee review under Brazilian National Health Council Resolution 510/2016 and CONEP Circular Letter 17/2022. Counts below five are suppressed at municipal level, and no re-identification is attempted.

COMPETING INTERESTS AND FUNDING

The author declares no competing interests. This work received no funding, and the project accepts no resources from manufacturers of technologies it assesses or comments on. This text has not been peer reviewed, reviewed by a health professional or subjected to independent human validation of the research steps carried out with AI (the author reviews only the final text), and should be read as a working note. Corrections create a new version, and the previous one remains accessible.

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